

SPRINGFIELD LINCOLN, IL, USA

WMO: 724390

Lat: 39.845N

Lon: 89.684W

Elev: 181

StdP: 99.17

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 93822

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.5	-14.4	-22.2	0.5	-16.3	-19.5	0.7	-13.8	12.3	1.7	11.4	0.8	3.6	280	0.522

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.8	33.8	24.9	32.6	24.4	31.3	23.6	26.6	32.0	25.7	30.8	24.8	29.6	3.9	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.0	20.5	30.3	24.1	19.4	29.2	23.3	18.5	28.1	83.8	31.9	79.8	30.9	76.1	29.6	29.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.0	9.5	8.5	-22.0	35.8	3.7	1.5	-24.7	36.9	-26.8	37.8	-28.9	38.6	-31.6	39.7		
(5)			-22.4	27.9	3.6	0.7	-25.0	28.3	-27.1	28.8	-29.1	29.1	-31.7	29.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	12.1	-2.6	0.0	6.0	12.4	18.2	22.9	24.5	23.6	19.9	13.3	6.3	0.3		
	DBStd	10.82	6.90	6.60	6.31	5.27	4.77	3.59	3.15	2.99	4.39	5.37	5.69	6.11		
	HDD10.0	1341	392	286	158	33	2	0	0	0	1	27	138	304		
	HDD18.3	2936	649	514	385	189	65	7	1	2	33	174	360	558		
	CDD10.0	2119	2	5	33	105	256	387	449	423	298	129	28	4		
	CDD18.3	671	0	0	2	10	60	144	191	166	81	17	1	0		
	CDH23.3	6549	0	0	15	109	539	1383	2005	1538	802	153	4	0		
	CDH26.7	2423	0	0	1	21	161	522	816	571	295	36	0	0		
(14) Wind	WSAvg	4.0	4.8	4.8	4.9	5.1	4.1	3.4	2.8	2.7	2.9	3.8	4.5	4.5		
(15) Precipitation	PrecAvg	900	40	43	76	97	93	96	94	84	80	70	68	64		
	PrecMax	1339	114	124	200	252	225	234	273	213	218	163	176	227		
	PrecMin	643	1	12	25	18	13	16	23	12	0	4	6	14		
	PrecStd	171	28	24	37	56	51	58	56	48	54	42	40	43		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.6	19.4	25.3	29.2	32.2	34.3	36.6	34.9	34.1	30.1	23.4	18.0		
		MCWB	13.4	14.0	17.0	18.4	22.5	24.3	25.5	24.6	23.3	20.6	15.7	15.2		
	2%	DB	13.1	15.9	21.9	26.6	30.3	32.9	33.9	33.2	32.2	27.3	20.0	14.6		
		MCWB	11.1	11.9	15.2	17.1	21.0	24.0	26.1	24.8	22.5	18.9	14.5	11.7		
	5%	DB	9.8	12.7	18.9	24.0	28.5	31.5	32.4	31.6	30.1	24.7	17.5	11.6		
		MCWB	7.4	9.3	13.6	16.1	20.0	23.5	25.3	24.0	21.4	17.6	13.5	9.3		
	10%	DB	6.4	9.5	15.9	21.5	26.6	29.8	30.9	29.9	28.1	22.1	15.2	9.0		
		MCWB	4.3	6.5	11.7	14.9	19.1	22.6	24.4	23.3	20.3	16.6	11.7	6.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.5	15.4	18.3	20.4	24.0	26.5	27.9	27.3	25.3	22.7	17.7	15.7		
		MCDB	16.2	17.9	23.6	26.7	30.3	31.8	33.3	32.7	31.9	27.3	21.0	17.7		
	2%	WB	11.4	12.8	16.3	18.7	22.7	25.4	26.9	26.1	24.0	20.9	15.7	12.6		
		MCDB	12.8	15.1	21.0	23.7	28.1	30.9	32.3	31.6	29.7	24.7	18.7	14.2		
	5%	WB	7.6	9.6	14.4	17.4	21.5	24.6	26.1	25.1	22.8	19.3	14.1	9.5		
		MCDB	9.6	12.6	18.3	22.0	26.6	29.8	31.3	29.8	27.7	22.9	16.8	11.3		
	10%	WB	4.4	6.5	11.9	15.9	20.4	23.6	25.2	24.1	21.7	17.5	11.9	6.7		
		MCDB	6.3	9.4	15.2	20.6	24.9	28.4	29.9	28.3	26.2	21.6	15.0	8.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.5	9.1	10.4	11.7	11.1	10.7	10.8	10.9	12.9	11.8	9.8	8.0		
		MCDBR	12.0	13.1	13.5	14.5	12.6	11.9	11.5	11.9	13.8	13.9	12.6	11.0		
		MCWBR	9.8	9.9	8.3	7.9	5.7	5.2	5.3	5.3	5.8	7.1	8.6	8.9		
		MCDBR	11.5	12.4	12.1	12.3	11.1	10.5	10.5	10.2	11.7	11.3	11.2	10.0		
	5% WB	MCWBR	10.0	10.1	8.1	8.1	5.9	5.2	5.4	5.2	5.5	6.8	8.9	9.0		
(40) Clear-Sky Solar Irradiance	taub		0.298	0.312	0.343	0.389	0.428	0.432	0.458	0.445	0.405	0.356	0.306	0.294		
	taud		2.451	2.404	2.397	2.271	2.221	2.249	2.163	2.192	2.281	2.423	2.510	2.477		
	Ebn at Noon		872	908	911	885	854	848	821	822	833	844	857	849		
	Edh at Noon		79	96	108	130	140	137	148	140	120	93	75	72		
(44) All-Sky Solar Radiation	RadAvg		1.88	2.73	3.69	4.84	5.51	6.29	6.27	5.53	4.70	3.24	2.17	1.58		
	RadStd		0.22	0.25	0.27	0.33	0.44	0.48	0.39	0.35	0.33	0.37	0.20	0.20		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB		1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+134	+99
(47) Regional (40 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+106	+75

Nomenclature: See separate page